

Deauthentication Attack in IEEE 802.11 Networks

Kevin Shaun Dsouza
Dept of Information Technology
Don Bosco Institute of Technology
Mumbai, India
kevin_dsouza@outlook.in

Disha Hemant Sanil
Dept of Information Technology
Don Bosco Institute of Technology
Mumbai, India
sanildisha3@gmail.com

Kuljeet Singh Bhengura
Dept of Information Technology
Don Bosco Institute of Technology
Mumbai, India
kbhengura@gmail.com

Prasad J Padalkar
H.o.D, Information Technology
Don Bosco Institute of Technology
Mumbai, India
prasad.dbit@dbclmumbai.org

Abstract—Wireless networks are unlike wired when it comes to security factor, as in they are considered fundamentally insecure as they transmit the data via radio waves. This project highlights the weak security factor of the MAC layer. The objective of this project is to investigate a special Denial of Service(DoS) attack against 802.11 wireless networks. This attack is known as the Deauthentication Attack which is launched against the 802.11 based wireless networks. When a client needs to disconnect from the wireless access point, it sends special frames known as deauthentication or disassociation frames. Being unencrypted, these frames do not require an authenticated user. Hence, an attacker can craft these frames and send them to clients of an access point, such that the client assumes the frames to be legitimate and coming from the access point and not the attacker. In this project, an efficient and economic solution is proposed to detect and prevent Deauthentication Attacks by verifying deauthentication frames.

Keywords—Wireless networks, MAC Layer, IEEE 802.11 Deauthentication, access-point

I. INTRODUCTION

In these recent years, the ubiquitous networks around us have advanced from a simple means of communication to a ubiquitous computational infrastructure. Networks have become larger, faster, and highly dynamic. IEEE 802.11-based networks have been very efficient and successful as they only need inexpensive hardware devices that operate on a free spectrum with economic deployment. Due to their popularity and usage, 802.11 networks consequently became the target of a vast variety of attacks. Researchers and industrial companies have been trying to fix these vulnerabilities in IEEE 802.11 networks by providing a number of standards and protocols. However, some defects can still not be outplayed by any of these protocols, one of which is the deauthentication attack described as follows.

In 802.11 there are mainly 3 types of frames: Data, Control and Management Frames. The Data and Control frames are both authenticated and encrypted but the Management frames are neither authenticated nor encrypted. So, this leads to the Management frames to be easily spoofed and duplicated by an attacker. Hence, the attacker can perform various DoS attacks. There are various types of Control frame present such as the Acknowledgement (ACK) frame, Request to Send (RTS) frame and Management frames as Beacon and Probe Request/Response, Authentication, Association Request/Response Frames, Deauthentication / Disassociation frame which is easily spoofed by the attacker [1]. The basic working and functioning of the Management Frame are shown in Figure 1.

Wireless Local Area Network (WLAN) security attacks are majorly classified into two categories: Crypto attacks and Denial of Service (DoS) attacks. The former includes unauthorized access, man-in-the-middle, masquerading, eavesdropping, replay, session hijacking attacks. In general, security solutions of VPN and 802.11i [2] provide an efficient and economic way of handling Crypto attacks. Unfortunately, these solutions do not protect WLANs against DoS attacks. Major DoS attacks on WLANs include authentication request flooding (AuthRF), association request flooding (AssRF), deauthentication flooding and so on. These DoS attacks render the WLAN and also some of its wireless nodes, out of service.

The absence of any kind of encryption at both of control and management frames exposes the network to consistent DoS attacks. DoS attacks may target different layers of the Open System Interconnection (OSI) model. The Application layer, Transport layer, Network layer, Link layer and Physical layer are all vulnerable to DoS attacks. Attackers can launch DoS attacks by flooding deauthentication and disassociation frames. These attacks are very effective because the target access point will be unable to serve a legitimate client. This paper only focuses on the Deauthentication/Disassociation attacks on the Media Access Control Layer.

This survey paper is organized as follows: in section II we discussed related work, section III we mainly consider various types DoS Attacks on Layer 2 of OSI model, section IV Experimental Analysis of DOS, section V solutions provided to DOS and finally in section VI we conclude.

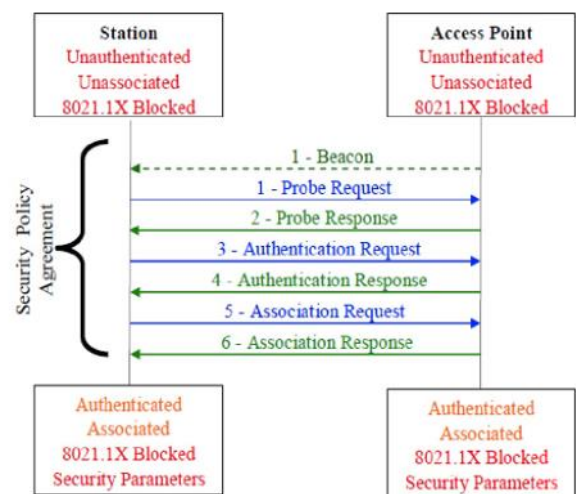


Figure 1: Working of the Management Frames

II. RELATED WORK

A great deal of research works have been analyzed on the security and different types of attacks on 802.11 networks.

Before we can finally dive into our survey of Wi-Fi Deauthentication and Deauthentication Detector approaches per se, we must take a moment to make a few much-needed clarifications.

- I) *Clarification 1:* Our goal in this section is to introduce, organize and review the existing systems similar to our systems, rather than to detail the exact algorithms that those systems employ. This is because common trends in the algorithmics of the reviewed solutions are currently rather limited across different sensing modalities.
- II) *Clarification 2:* The second clarification we must make is that since the authors of the solutions reviewed in this section often do not agree on common performance metrics or even experimental scenarios, we are forced to compare different approaches in rather qualitative terms. Thus, we use words such as “accuracy” and “precision” loosely to denote a measure of the average error and a measure of classification correctness.

A. A Survey of Denial of Service Attacks and its Countermeasures on Wireless Network [3]

This paper by Dr. L. Arockiam Lawrence makes a survey on different types of DoS attacks and various countermeasures provided for networks based on Access Points (AP). In this paper, the primary attacks studied are called the Deauthentication and Disassociation Flooding DoS attacks. These attacks are studied through experiments and the most probable solutions are listed. This paper also discusses and tries to explain the vulnerabilities due to rogue access point, null data frames and Access Points.

B. Fragility of the Robust Security Network: 802.11 Denial of Service [4]

This paper highlights two other DoS vulnerabilities: one vulnerability currently in draft 802.11w implementations discovered by IEEE 802.11 TGw, and a new vulnerability which is present in 802.11, which is still present in the 802.11w improvement. Attacks which exploit the first vulnerability are significantly more efficient than any known 802.11 DoS attacks, while attacks which exploits the second vulnerability have the efficiency and practicality equivalent to a disassociation attack. This paper then provides an experimental analysis of these attacks, demonstrating their feasibility using freely available software and cheap and compact hardware. Finally, the primary cause of these vulnerabilities occurring in the networks is discussed and an efficient backward compatible solution is proposed.

C. Access points vulnerabilities to DoS attacks in 802.11 networks [5]

The primary objective of this research paper was to investigate how DoS attacks can be carried out. In particular we tried to identify some simple DoS attack stream flows that might lead to a DoS effect and then observed the reactions of various types of infrastructure networks to such attacks. The summary of the conclusion is given as follows:

- i) The 802.11 protocols fundamentals rely on the exchange of request/response messages: each request sent by a station (STA) in the network triggers a corresponding response from its client/s, which can be, in turn, another station or an Access Point (AP) [8];
- ii) Infrastructured networks are based on an access point (or a collection of them) acting like a central node through which every communication is routed, thus an access point can easily become a bottleneck (congestion) for the entire network (or, at least, for the Basic Service Set it establishes).
- iii) Attack patterns should be as straight forward as possible, in order to apply to both, Open networks and WEP-protected networks. From this perspective, a malicious station (like the one we are using for our experiment) should be able to launch an attack even if it is neither authenticated nor associated with the target network.

D. MAC-layer denial of service attacks in IEEE 802.11 ad hoc networks: analysis and counter measures [6]

In this paper, the authors, Yihong Zhou, Dapeng Wu and Scott M. Nettles discuss two types of distinct MAC layer DoS attacks that can be easily carried out by any person, not even a hacker. These attacks are basically:

- i) Attacks launched from a single attacking device by injecting massive data flows into the network and;
- ii) Attacks launched from two colluding adversaries by sending massive data flows directly to each other.

As a countermeasure and to protect networks from the first attack, they propose a packet-by-packet authentication scheme so that the legitimate nodes have the ability to reject data transmission requests from unauthenticated adversaries. To counter the second category of the attack, they establish several counter measures such as a fair MAC protocol, using protecting traffic flows along with distance adjustment. The influence of this paper is that it reveals the attacks that could be launched by any person, which is a potential threat to MANETs.

E. MAC Layer DoS Attacks in IEEE 802.11 Networks [7]

In this paper the authors discuss and explain DoS attacks which exploit the MAC layer vulnerabilities of IEEE 802.11 networks. In recent years, DoS attacks in wireless networks have been getting the attention of researchers and it has been demonstrated that DoS attacks targeted at the MAC Layer can easily be deployed using cheap, off the shelf equipment. In most cases the attacking device replicates the MAC addresses of wireless devices as its own in order to halt the operation or

deny service of the wireless network. MAC address spoofing can be easily done because the IEEE 802.11 standard does not provide per frame source authentication for neither the control nor the management frames. Even the new amendment to the WLAN security standard IEEE 802.11i does not overcome these problems. Many software tools are easily obtainable for attackers to launch such types of attack. In this paper, the authors classify MAC layer DoS attacks into three categories, and compare the existing countermeasures to such attacks.

III. VARIOUS TYPES OF DOS ATTACKS

A. Beacon frames DOS Attack

Beacon frames always send by the legitimate AP in the network to exhibit its presence when an attacker spoofs the mac address of an AP for making rouge AP in that network. After which both legitimate and rouge AP is sending beacon frames, and the clients are attached to the rouge AP acting as its original AP.

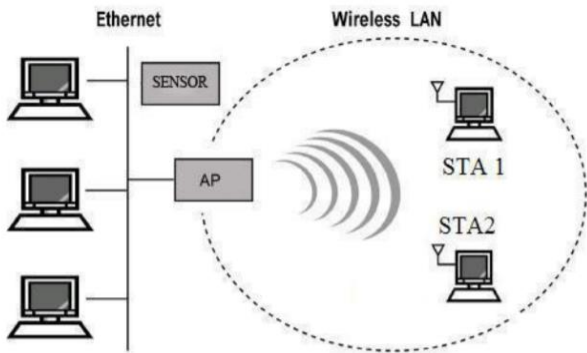


Figure 2: Working of the Beacon frames DoS Attack [31]

B. Probe Request /Response DOS Attack

The probe request message is sent by the client when it wants to join any AP and the APs present in that network answer to that client as a probe response message. The attacker performs probe request DOS attack by continuously changing its mac address to fool the AP as there are various clients in the network which are waiting to connect to them. This uses all of the network resources of that AP, and neither legitimate nor fake clients can access that AP. This can also be done by making a rouge access point.

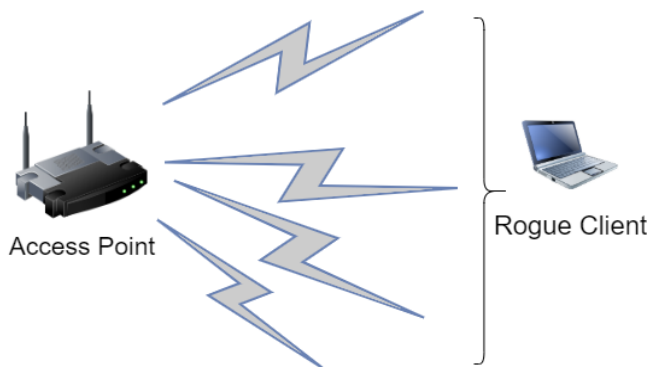


Figure 3: Working of the Probe frames DoS Attack

C. Authentication / Association DOS Attack

The client sends an Authentication / Association request, which is responded by AP as an authentication/association response. An attacker spoofs the mac addresses of the clients or arbitrarily changes his system mac address and sends the authentication/association requests to the AP. The AP then sends the corresponding authentication/association responses, and as AP is busy handling such forged authentication/association requests or response messages, it cannot respond to the original requests, and hence DOS attack takes place.



Figure 4: Authentication/Association Flood DoS Attack [33]

D. Deauthentication / Disassociation DOS Attack

Deauthentication / Disassociation attack is a very risky attack because these dismiss the original connections. This attack is performed by the attacker by either spoofing the mac address of the client or mac address of the AP.

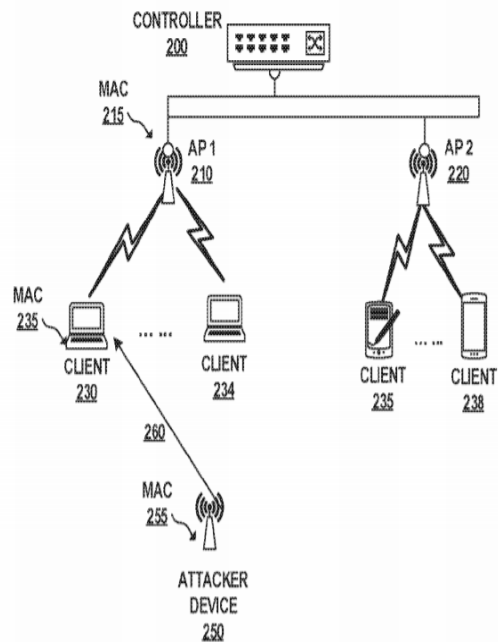


Figure 5: Working of the Deauthentication Attack [32]

TYPES OF DOS ATTACKS AND COUNTERMEASURES

Attack	Target	Existing countermeasures
Probe Request Attack	AP	Signal Print
Authentication Request Attack	AP	Signal Print, Client Puzzle
Deauthentication Attack	Station and AP	Signal Print, MAC Spoof Detection, Delaying the effects of request
Association Request Flood	AP	Signal Print
Deassociation Attack	Station and AP	Signal Print, MAC Spoof Detection, Delaying the effects of request
Virtual Carrier Sense Attacks	Medium Access	Explainability of Collision, Spatial Retreats
Sleeping Node Attack	Station and AP	Limiting Duration Field Value, Signal Print, MAC Spoof Detection

Figure 6: Deauther Module [7]

IV. EXPERMENTIAL ANALYSIS OF DEAUTHENTICATION ATTACK

In this demonstration we are mainly focused upon the Deauthentication/Disassociation attack. We created a module using a cheap and off the shelf, ESP8266 board call the NodeMCU. We also integrated the setup with an on-board OLED screen for offline control. The experimental setup of our Deauther module is shown in figure 7 and 8.

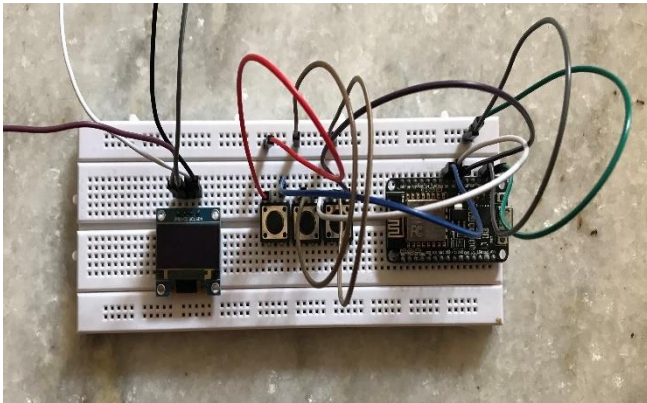


Figure 7: Deauther Module with add-on OLED Display

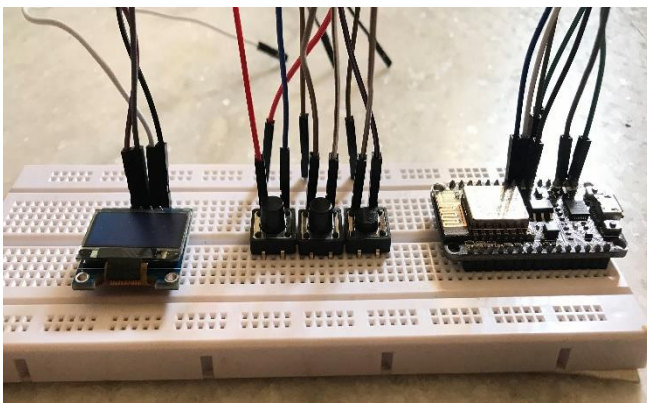


Figure 8: Deauther Module

The deauthentication packets are legitimately sent by the access points (AP) to the clients when the AP is overloaded or has any other operating system problems as shown in table 1. In such cases, it broadcasts deauthentication packets to all its connected client devices requesting them to disconnect from the network [29]. In our experiment, we set up our device to spoof/duplicate the MAC address of a nearby AP and then start broadcasting deauthentication packets. Hence, all the client devices will receive the deauthentication packets and be fooled that it is from the legitimate source and, therefore, disconnect from the network as shown in fig 1.

Reason Code	Description
Code-0	Reserved
Code-1	Unspecified
Code-2	Prior authentication was not valid
Code-3	Station has left the basic service area and is deauthenticated
Code-4	Inactivity timer expired and station was disassociated
Code-5	Disassociated due to insufficient resources
Code-6	Incorrect frame type or subtype received from unauthenticated station
Code-7	Incorrect frame type or subtype received from unassociated station
Code-8	Station has left the basic service area and is disassociated
Code-9	Association or reassociation requested before authentication is complete
Code-10 to 65535	Reserved*

Table 1: Various reasons for Deauthentication broadcasts

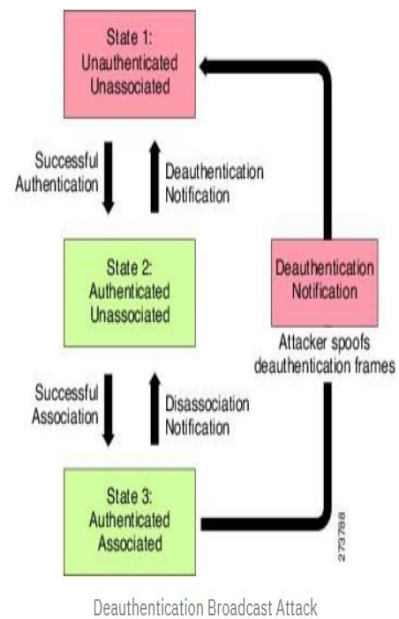


Figure 9: Deauthentication Attack process [30]

We should also keep in mind that what we are performing is a deauthentication attack and not a jamming attack. Jamming is something that interfered or plays with a particular frequency to disrupt all communications on that frequency, whereas, our Deauther is a module that broadcast deauthentication packets into the channels and request devices to disconnect. A detailed differentiation is given in figure 10.

Wifi Jammer vs Team 17 Deauther	
WIFI JAMMER	TEAM 17'S DEAUTHER
Creates noise on a specific frequency range (i.e. 2.4GHz)	Exploits a particular vulnerability of IEEE 802.11 Standards
Targets Frequency Bands . Eg: 2.4GHz or 5GHz Band	Targets particular Access Points only
All Wifi's on the same frequency get jammed	All Wifi's do not get affected. Only the Target AP and Clients are affected
Mostly ILLEGAL to use	Mostly LEGAL to use
Increases the Noise:Signal Ratio. Therefore, AP is not ACCESSIBLE	Does not affect the Noise:Signal Ratio. AP is reachable but Client thinks AP sends a Deauth Signal
EXPENSIVE	MUCH CHEAPER

Figure 11: Jammer vs Deauther

To practically carry out this deauthentication/ disassociation DoS attack following steps are to be performed for:

- i) Firstly, we have to power up the NodeMCU with an USB cable.
- ii) Open your Wi-Fi settings and find for a network named “pwned”. We then connect to this network using the password we have programmed it with.

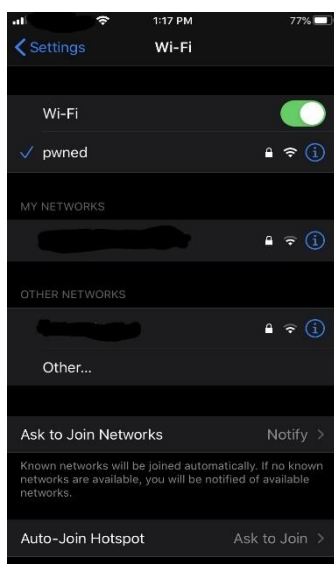


Figure 10: Connecting to the network

- iii) We open our web browser and enter the IP: 192.168.4.1. This is the IP on which the NodeMCU hosts its webpage.

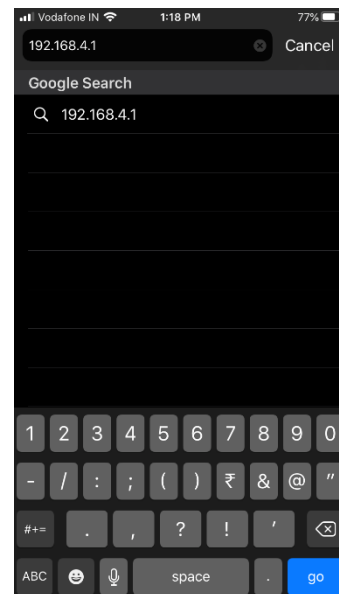


Figure 11: Entering IP address

- iv) On this webpage we choose the option to scan the nearby access point. Once, the access points are scanned we get a list of the nearby access points (AP). We can choose which AP we want to attack and proceed.

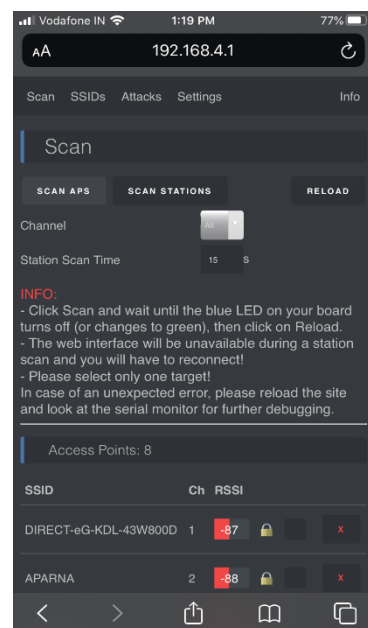


Figure 12: Web Page hosted on 192.168.4.1

- v) Once we choose the AP, we go to the “Attacks” webpage and choose an attack; Deauthentication, Probe or Beacon Frames attack.

- vi) Once we choose the type of attack, we press the start button and wait for the SoC to do its job.

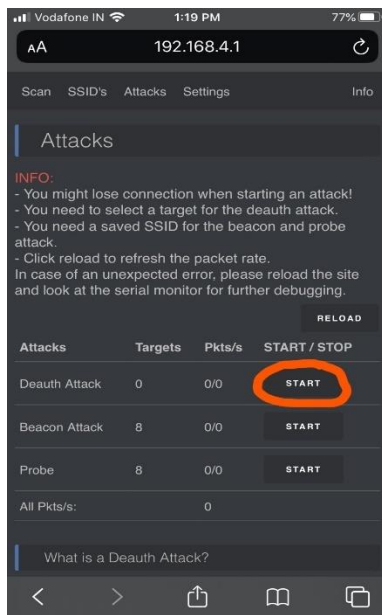


Figure 13: "START" an attack

vii) After a while, if you want to stop the attack press the stop button and the SoC stops executing further attacks.

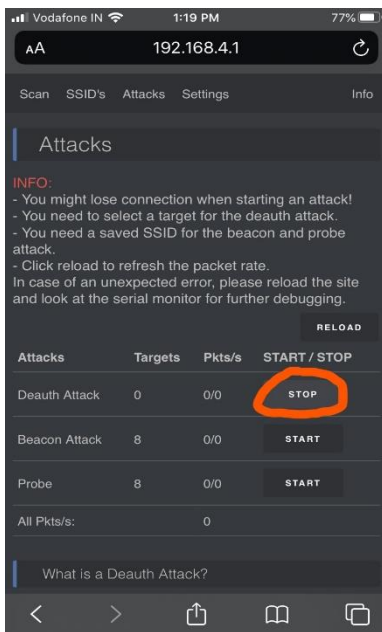


Figure 14: "STOP" an attack

V. SOLUTIONS PROVIDED FOR DOS ATTACKS

There are various solutions available such as:

A. Pseudo Random Sequence Number Solution

A sequence number solution is provided for disassociation DOS attack, which is the highly serious attack. The primary idea was to use a pseudo randomised sequence number for a disassociation or a deauthentication packet acknowledgement rather than sequential sequence number. The receiving side will confirm authenticity of frame by checking whether the pseudo randomized sequence number is within the acceptable sequence numbers range or not. If not, the frame will be discarded, and if yes, processed. The solution will need a firmware upgrade [24]

B. Authentication Based on Pseudo Randomized Technique

This technique is mainly based on a modified Pseudo Random Number authentication mechanism that can be applied to prevent all above-mentioned DOS attacks on 802.11 based WLANs. The Frame Control and Frame Check Sequence (FCS) are the only common fields present amongst all the management frame. The frame control field contains basic information in its subfields and FCS field which contains the 32-bit Cyclic Redundancy Code (CRC) which is error checking code. The FCS field in the mac frame format is the only flexible common field which may be utilized for supporting the authentication mechanism using pseudo random numbers for all management frames. The replacement of CRC32 bit using CRC16 would leave 16 bits in the FCS field which will be used for inserting the pseudo random number. It is also recommended by the author that this solution could be quite a robust technique to prevent deauthentication, disassociation-based attacks.[24].

C. IDMDOS algo

The process occurs in 5 steps:

- 1) It takes mac address of the client.
- 2) Check in the intruder table if exist ignore the request
- 3) Else if mac address in the authentication client table if exist, ignore the request & store mac in intruder table.
- 4) Else if check in client table if exist, ignore it.
- 5) Else accept the request & start communication [25].

D. Letter envelop Protocol-Light weight solution to WLAN Denial of Service attacks

The solution that is proposed in this research paper is like an amendment or improvement on the current 802.11-based protocol. To prevent the deauthentication or disassociation attack, the author, Thuc Nguyen et al, uses letter-envelop protocol to authenticate management frames in association process. After authentication process between wireless client and a legitimate access point, the association takes place. Similar to attacking on the original 802.11 protocol, they do the same flooding and DoS attack on proposed implementation system. The system can prevent these attack types.

E. MAC Address Spoof Detection

One further method to detect MAC address spoofing is based on the sequence number field. This value is incremented by one for each non-fragmented frame. An attacker does not have the ability to alter the value of this sequence number as they generally can't control the firmware functionality of their wireless card [26, 27]. Through the analysis of sequence number patterns of the captured wireless traffic, detection systems were shown to be capable of detecting MAC address spoofing to identify deauthentication/disassociation attacks [27]. Sequence number-based MAC address spoof detection systems are also valuable until reverse-engineered "attack cards" allowing frames with arbitrary sequence numbers become common place [27,28].

VI. CONCLUSION

Various DoS attacks on Wireless Local Area Networks (WLAN) remain possible because these networks lack reliable and robust authentication mechanisms for control and management frames before upper-layer authentication mechanisms are evoked and user credentials are securely established. In this research paper we discussed one such type of attack on IEEE 802.11 networks called the Deauthentication attack. As understanding the attack would be difficult theoretically, our team carried out attacks practically and tried our best to study and discuss different tactics or strategies used for defending the Wi-Fi against the Deauthentication Attack. Based on the practical experiments that were basically held in our homed lab, the necessity for automated detection of deauthentication attacks method raised alongside a patch to mitigate both deauthentication and disassociation attacks. Our proposed technique for deploying a deauthentication attack is lightweight, cheap and can easily be deployed to both open as well as encrypted networks. Demonstrating a deauthentication attack and provide various points of differentiations between deauthentication and a jamming attack. Comprehending that forensics is an important part of Cybersecurity we planned on adding a Packet-Sniffer. The Packet Sniffer sniffs all the packets and network activity going on nearby and saves all the packets as pcap files. After a Deauthorisation or Disassociation attack has been detected the client(s) can use these pcap files to understand who and from where was the Deauth attack performed. Hence, automated detection mechanism was implemented successfully and proved its efficiency well.

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